

Taste-Based Social Discovery Platform

Project Proposal for UCS503P

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August 20, 2026

1 Higher-order goal

... secondary framing

This project aims to improve social discovery by helping users find and connect with people who share similar tastes across multiple domains of entertainment and interest, rather than a single category such as music alone. While the broader ambition is to rethink how people find community online, the immediate engineering objective is to translate *taste similarity* into a measurable, explainable compatibility score delivered through a working software system.

2 Time-to-value

... primary heuristic

- We will deliver meaningful increments quickly by prototyping the core taste-matching workflow (profile → Taste DNA → compatibility score) and validating it with real users within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations, starting with a simple weighted-similarity matching algorithm before attempting any NLP/embedding-based approach.
- Success is defined by what can be delivered and measured in the first iteration – a working “Find My People” flow with an explainable match score – then improved based on user feedback.

3 Problem Statement

Existing social platforms are typically organized around a single content category (e.g., music, photos, short video) or around a follower-based model that optimizes for reach rather than genuine compatibility. This creates several pain points for users who want to find people who actually share their taste:

- **Fragmentation:** a person’s taste is spread across separate apps – one for music, one for movies, one for books – with no unified representation of “who they are.”

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- **Opaque recommendations:** existing “suggested for you” features rarely explain *why* two users were matched, reducing trust in the suggestion.
- **Follower-centric design:** following someone is a one-directional, low-commitment action that does not guarantee any shared interest, leading to noisy, low-relevance feeds.
- **Weak niche discovery:** users with specific or unusual combinations of interests (e.g., dark academia aesthetics, psychological thrillers, and Formula 1) struggle to find others who share that exact combination.

Why this matters now (contextual relevance): As content categories multiply and attention fragments across apps, a unifying, cross-domain notion of “taste” is increasingly useful for meaningful (rather than purely algorithmic) social discovery.

Positioning relative to existing tools: single-category platforms (e.g., Letterboxd for film, Goodreads for books) already let users track taste within one domain, and dating/social apps increasingly surface shared interest tags as a matching signal. What this project adds is not a new content category, but (i) a *cross-domain* taste representation spanning multiple interest types at once, and (ii) an *explainable* compatibility score that shows the specific shared interests behind a match, rather than an opaque “suggested for you” ranking.

4 Proposed Solution

... *Software Proposition*

4.1 Overview

Build a **mobile-first** “Taste Discovery” platform with the following components:

- **Taste DNA profile:** users build an interest profile from keywords spanning music, movies, and books at launch, with additional categories (memes, games, shows, sports, food, photography, custom/niche tags) added in later iterations.
- **Content Drops:** lightweight posts through which users share and tag content they like, categorized by domain.
- **Taste-based matching engine:** computes a per-category and overall compatibility score between users based on shared keywords and content.
- **Find My People:** a ranked, explainable list of recommended users, each annotated with the specific shared interests driving the match.
- **Connection system:** opt-in, bidirectional connections (request/accept) rather than one-directional following, gating private content behind mutual connection.
- **Communities:** niche, tag-based groups (e.g., #DarkAcademia, #F1Girls) for users to congregate around specific shared interests – deferred to a subsequent iteration.

4.2 Core workflow

1. User signs up, creates a profile, and selects interest keywords to build their Taste DNA.
2. User explores and engages with content across the launch categories (music, movies, books) through likes, saves, and Drops.
3. System computes per-category and overall compatibility scores against other users.
4. User opens “Find My People” and views ranked matches with an explainable “why you matched” breakdown.
5. User sends a connection request; once accepted, the match’s full profile and interest details unlock.

4.3 Operational constraints for an educational setting

- Keep the matching algorithm computationally lightweight (weighted keyword/content overlap) for the MVP; avoid dependence on research-grade embedding models initially.
- Design for deployability using common managed backend/database hosting and free-tier external content APIs (TMDB, Spotify, Google Books).
- Keep authentication, storage, and moderation simple but secure by design from the first iteration, since the platform handles personal interest and chat data.

5 Solution Approach

... Engineering focus

This is an engineering course project, so we focus on a correct, secure, and scalable matching-and-connection workflow rather than novel recommendation-system research.

5.1 Taste-matching

... non-research

The MVP matching algorithm will be heuristic-based:

- Represent each user's Taste DNA as a set/vector of weighted keywords across categories.
- Compute per-category similarity (e.g., Jaccard or cosine overlap on keyword sets) and combine into a weighted overall compatibility score, without claiming state-of-the-art accuracy.
- Always surface the specific shared keywords/content behind a match, so recommendations are explainable rather than opaque.

5.2 System architecture

... mobile/web-based solution

A typical mobile-client / backend-API / database architecture:

- **Frontend:** Flutter mobile client for profile, explore, matching, chat, and community screens.
- **Backend API:** FastAPI service handling authentication, profile/Taste DNA management, matching computation, connections, and chat metadata.
- **Database:** MongoDB storing user profiles, keywords, content, Drops, connections, communities, and reports.

5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run backend/frontend tests on every change.
- Produce repeatable deployment artifacts to a staging environment.
- Enable safe, frequent releases of incremental matching and social features.

6 Evaluation Criterion

... measurable and attributable

We define success using measurable metrics tied to the core matching-and-connection workflow.

6.1 Primary evaluation metric

Match-to-connection rate (MCR): the percentage of “Find My People” recommendations that result in a sent connection request within a defined window.

- Target: $\text{MCR} \geq 20\%$ during the pilot window.
- Attribution: measured from database timestamps comparing recommendation impressions to connection-request events.

6.2 Secondary evaluation metrics

- **Connection acceptance rate:** percentage of sent requests that are accepted, as a proxy for match quality.
- **Explainability satisfaction:** pilot users rate the “why you matched” explanation on a simple 1–5 clarity/usefulness scale.
- **Taste DNA completion rate:** percentage of new users who complete a minimum viable Taste DNA (e.g., ≥ 5 keywords across ≥ 2 categories) during onboarding.
- **Match explanation density:** average number of shared-interest tags surfaced per top-5 recommended match. This is measurable directly from stored data without waiting on user behavior, and serves as an early proxy for match quality even before enough connection events accumulate.
- **Reliability:** availability of staging/production for login, profile, and matching endpoints (e.g., $\geq 99\%$ uptime in pilot).

6.3 Pilot validation plan

... high-level

- Recruit 20–30 pilot users with varied interest profiles to seed a meaningful matching pool.
- To mitigate the cold-start problem inherent to a small pilot, supplement real signups with a curated set of synthetic profiles carrying deliberately overlapping taste keywords, ensuring the matching engine has enough density to produce meaningful, demoable matches even before organic adoption grows.
- Compare MCR and acceptance rate before/after refining the weighting of the similarity algorithm within the iteration window.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in number of users and matching computations by:
 - decoupling the matching engine from the core API for independent scaling,
 - precomputing/caching compatibility scores instead of recomputing on every request,
 - using indexed queries on keyword and category fields.
2. **Operationally deployable (practically):** The system should run on common hosting:
 - managed document database (MongoDB Atlas),
 - containerized or platform-hosted backend deployment,
 - CI/CD pipeline for staging/production releases.

8 Engine availability heuristic

A mature set of “engines” (tooling/services) is available to implement this as a mobile/web project:

- Mobile framework (Flutter) for cross-platform client development.
- Web framework (FastAPI) for REST APIs.
- Managed document database (MongoDB Atlas).
- Authentication service (Firebase Authentication).
- Object storage for images/memes (Cloudinary).
- Established similarity/matching libraries (NumPy, scikit-learn) for the MVP algorithm.
- External content APIs (TMDB, Spotify, Google Books) to avoid building content catalogs from scratch.
- Test frameworks (Pytest, Flutter test) and CI runners for staging deployment.

We will use these mature components to focus on matching-workflow design, explainability, and security rather than inventing new infrastructure.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration (MVP)

- User authentication and profile creation
- Taste DNA keyword selection across music, movies, and books
- Content Drops: create, view, like, and save
- Weighted-similarity compatibility scoring between users
- “Find My People” ranked list with explainable match reasons
- Connection requests: send, accept, reject
- Privacy settings (public/private profile, hide/show interests)
- Block/report functionality
- CI: build + backend unit tests + linting; CD to staging

The MVP deliberately excludes chat, memes, and communities so that the first iteration’s demo stays centered on the core loop: building a Taste DNA, computing an explainable match, and connecting.

9.2 Subsequent deliverables

- Basic one-to-one chat between connected users
- Meme category: upload, tag, and share
- Niche communities: create/join, community posts and discussions
- Admin/moderator dashboard for reports and content moderation
- Notification layer for connections, messages, likes, and new matches
- Semantic/embedding-based matching (Sentence Transformers, FAISS) as a research-grade upgrade to the MVP keyword-overlap algorithm
- Vibe Rooms (temporary interest-based real-time spaces) and group chats

10 Risks and mitigations

- **Data privacy / consent:** minimize collected personal data; keep interest visibility opt-in; never store plaintext passwords; keep secrets in environment variables, not the frontend.
- **Cold-start matching:** with few pilot users, compatibility scores may be sparse or low-quality; mitigate by seeding a diverse pilot cohort and lowering the minimum-keyword threshold for early matches.
- **Content moderation risk:** user-uploaded profile images and Drops could surface inappropriate content, and this risk grows once memes and chat are added in later iterations; mitigate with upload validation, reporting flows, and admin moderation built in from the first iteration rather than bolted on later.
- **Evaluation measurement ambiguity:** instrument recommendation-impression and connection-request events with clear timestamps before development begins.
- **Operational issues in pilot:** use staging early; ensure CI/CD produces repeatable deployments to reduce pilot downtime.

11 Summary

This project proposes a deployable taste-based social discovery platform that helps users find and connect with people who share similar interests across multiple content domains, rather than a single category or a follower-based model. It meets the course criteria by emphasizing time-to-value through rapid iterations starting with a lightweight, explainable matching algorithm, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially match-to-connection rate). It remains focused on engineering workflow, security, and scalability readiness in its MVP, while reserving semantic/embedding-based matching as a well-scoped future enhancement rather than a first-iteration research dependency.